

Abstract

In this note, we explore how a matroid structure on one part of a bipartite graph can induce a matroid on the other part, which is a generalization of transversal matroids known as Rado matroids. We derive the corresponding rank function, and connect these ideas to Rado's and Hall's theorems. Finally, we use this idea to characterize the matroid union.

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We begin with our main theorem:

Theorem 1. Given a bipartite graph Δ with disjoint vertex sets X and Y , if we have a matroid structure on Y , denoted as $\mathcal{M}_Y = (Y, \mathcal{I}_Y)$, then $\mathcal{M}_Y$, together with Δ , induces a matroid on X , denoted as $\Delta(\mathcal{M}_Y)$ ¹. Moreover, $A \in \mathcal{I}(\Delta(\mathcal{M}_Y))$ if and only if there exists $B \in \mathcal{I}_Y$ such that $\Delta[A \cup B]$ has a perfect matching. (Or, slightly informally, $A \subseteq X$ is independent in $\Delta(\mathcal{M}_Y)$ if and only if A can be matched to an independent set in $\mathcal{M}_Y$ via Δ .)

Moreover, for any $S \subseteq X$, the rank function is given by:

$$r_{\Delta(\mathcal{M}_Y)}(S) = \min_{T \subseteq S} \{r_{\mathcal{M}_Y}(N(T)) + |S - T|\}$$

We postpone the proof to later. For now, we naturally raise several questions:

- Q1: Why is it even possible to naturally have a matroid on the other side of a bipartite graph, given a matroid on one side? How is the rank function interpreted?
- Q2: What if we 'derive' twice, or even more? i.e., what can we say about $\Delta(\Delta(\dots\Delta(\mathcal{M}_Y)\dots))$?
- Q3: How is this related to Hall's Marriage Theorem, and more generally, Rado's Theorem?
- Q4: For what kinds of given matroids $\mathcal{M}_X, \mathcal{M}_Y$, can we construct a bipartite graph Δ on X, Y s.t. $\mathcal{M}_X = \Delta(\mathcal{M}_Y)$?

To answer Q1, we first take a look at some examples:

Example 1. Let $\mathcal{M}_Y$ be the free matroid on Y . Then the induced matroid $\Delta(\mathcal{M}_Y)$ on X is the standard transversal matroid. By this interpretation, we understand transversal matroid naturally. The direct proof of showing "transversal matroids are matroids" by Edmonds and Fulkerson (1965) is tricky. See [Oxl06] Thm 1.6.2.

Example 2. Let Δ be a perfect matching between X and Y . Then the induced matroid $\Delta(\mathcal{M}_Y)$ on X is just a copy of $\mathcal{M}_Y$.

Example 3. Let Δ be a complete bipartite graph. Then what is the induced matroid $\Delta(\mathcal{M}_Y)$ on X ? Graphically, the completeness of Δ provides the total symmetry on X . In other words, it

¹In general, there is no specific name for this matroid. We can just call it *Derived matroid*, or *Rado matroid*.

must be uniform. As long as the cardinality of an arbitrary $X' \subseteq X$ is less or equal to the rank of $\mathcal{M}_Y$, it is independent. Or Arithmetically, we have:

$$\begin{aligned} r_{\Delta(\mathcal{M}_Y)}(X) &= \min_{K \subseteq X} \{r_{\mathcal{M}_Y}(N(K)) + |X - K|\} \\ &= \min_{K \subseteq X} \{r_{\mathcal{M}_Y}(Y) + |X - K|\} \\ &= r_{\mathcal{M}_Y}(Y) \end{aligned}$$

Proposition 1. (Edmonds and Rota, 1966) Given any increasing submodular function $f :$

$$2^E \rightarrow \mathbb{Z}_{\geq 0}$$

f naturally induces a matroid $\mathcal{M} = (E, \mathcal{C})$, where the circuits $\mathcal{C}$ consist all subsets $C \subseteq E$ s.t. C is minimal in the sense of $|C| > f(C)$.

Proof. We verify the circuit elimination axiom. The other 2 conditions are trivially satisfied. Take $\forall C_1, C_2 \in \mathcal{C}$ and $\forall e \in C_1 \cap C_2$. Note that:

$$|C_i| > f(C_i) \geq f(C_i \setminus e) \geq |C_i| - 1$$

This forces:

$$f(C_i) = |C_i| - 1$$

Now, we want to show : $C_1 \cup C_2 \setminus e$ contains a circuit. Suppose the contrary, we have:

$$\begin{aligned} f(C_1 \cup C_2 \setminus e) &\geq |C_1 \cup C_2 \setminus e| \\ &= |C_1 \cup C_2| - 1 \\ &= |C_1| + |C_2| - |C_1 \cap C_2| - 1 \\ &= f(C_1) + 1 + f(C_2) + 1 - |C_1 \cap C_2| - 1 \\ &\geq f(C_1 \cap C_2) + f(C_1 \cup C_2) - f(C_1 \cap C_2) + 1 \\ &= f(C_1 \cup C_2) + 1 \\ &\geq f(C_1 \cup C_2 \setminus e) + 1 \end{aligned}$$

Contradiction. ■

Remark 1. Proposition 1. is true for any increasing submodular function f . But if, in addition, we impose $f(\emptyset) = 0$, then we can write down the rank function for the induced matroid $\mathcal{M}_f$. For the entire following discussion, we assume this condition.

We now know what are the circuits of $\mathcal{M}_f$. And how to characterize the independent sets? Note that a set I is independent if and only if it does not contain any circuit. Hence:

Corollary 2.

$$I \in \mathcal{I}_{\mathcal{M}_f} \Leftrightarrow \forall I' \subseteq I, f(I') \geq |I'|$$
■

Proposition 3. Let E be the ground set, and let f be an arbitrary increasing submodular function with $f(\emptyset) = 0$. Then, the rank function of $\mathcal{M}_f$ is given by:

$$r_{\mathcal{M}_f}(S) = \min_{T \subseteq S} \{f(T) + |S - T|\}$$

Proof. Prove it. ■

Remark 2. Compare this with the rank function in Theorem 1:

$$r_{\Delta(\mathcal{M}_Y)}(S) = \min_{T \subseteq S} \{r_{\mathcal{M}_Y}(N(T)) + |S - T|\}$$

We may guess that: $\Delta(\mathcal{M}_Y)$ is nothing but a matroid induced by a submodular function f where $f = r_{\mathcal{M}_Y} \circ N$. We first verify the submodularity. For any subsets $A, B \subseteq X$, we have:

$$\begin{aligned} r_{\mathcal{M}_Y}(N(A)) + r_{\mathcal{M}_Y}(N(B)) &\geq r_{\mathcal{M}_Y}(N(A \cup B)) + r_{\mathcal{M}_Y}(N(A) \cap N(B)) \\ &\geq r_{\mathcal{M}_Y}(N(A \cup B)) + r_{\mathcal{M}_Y}(N(A \cap B)) \end{aligned}$$

Then, in order to understand Theorem 1 (and to answer Q1), we need to verify that $\Delta(\mathcal{M}_Y)$ is indeed the matroid induced by $r_{\mathcal{M}_Y} \circ N$. Recall that a subset $C \subseteq X$ is a circuit if and only if C is minimal s.t.

$$|C| > r_{\mathcal{M}_Y}(N(C))$$

By Corollary 2, $I \subseteq X$ is independent if and only if for all $I' \subseteq I$,

$$r_{\mathcal{M}_Y}(N(I')) \geq |I'|$$

And this is exactly the condition when the Rado's theorem holds:

Theorem 2. (Rado, 1942) Given a bipartite graph $\Delta[X \cup Y]$, and a matroid structure on Y , denoted as $\mathcal{M}_Y$. X can be perfectly matched to some independent subset of Y if and only if $\forall T \subseteq X$,

$$r_{\mathcal{M}_Y}(N(T)) \geq |T|$$

Proof. $\Rightarrow$ is trivial. For $\Leftarrow$, If we admit Theorem 1, then this is very straightforward to see: The condition implies $\forall T \subseteq X$,

$$r_{\mathcal{M}_Y}(N(T)) + |X - T| \geq |X|$$

and hence by Theorem 1,

$$r_{\Delta(\mathcal{M}_Y)}(X) = \min_{T \subseteq X} \{r_{\mathcal{M}_Y}(N(T)) + |X - T|\} \geq |X|$$

which means X is independent in $\Delta(\mathcal{M}_Y)$. And hence by Theorem 1 again, X can be perfectly matched to some independent subset of Y . ■

Remark 3. This indicates that Theorem 1 is stronger or equal to theorem 2. Later we will see that actually they are equally powerful. i.e., we will use theorem 2 to deduce the rank function $r_{\Delta(\mathcal{M}_Y)}(\cdot)$. For now, we give another proof of theorem 2 without admitting Theorem 1.

Theorem 2 can be rewritten in this way: Given a bipartite graph $\Delta[X \cup Y]$, and a matroid structure on Y , denoted as $\mathcal{M}_Y$. Then X can be perfectly matched to $\{y_i\}_{i \in X} \subseteq Y$ s.t. $\forall K \subseteq X$,

$$r_{\mathcal{M}_Y}(\{y_i\}_{i \in K}) \geq |K| \tag{1}$$

if and only if $\forall K \subseteq X$,

$$r_{\mathcal{M}_Y}(N(K)) \geq |K| \tag{2}$$

Proof. Again, $\Rightarrow$ is trivial. For $\Leftarrow$, we use induction: Take an arbitrary $K \subseteq X$. Let $n = |K|$. And let $(A_1, \dots, A_n)$ denote the family of neighbors of elements in K . Now we know that the family $(A_1, \dots, A_n)$, i.e., $N(K) = \bigcup_{i=1}^n A_i$ satisfies (2). If every A_i is a singleton, then (1) is naturally satisfied. Therefore, we may assume that $|A_1| \geq 2$, and it suffices to show that $\exists y \in A_1$ s.t. $(A_1 \setminus y, \dots, A_n)$ also satisfies (2).

Suppose the contrary that $\forall y \in A_1$, $(A_1 \setminus y, \dots, A_n)$ does not satisfy (2). Since $|A_1| \geq 2$, we may take such $y_1, y_2 \in A_1$. This means for both $i = 1, 2, \exists 1 \in K_i \subseteq X$ s.t.

$$r_{\mathcal{M}_Y}(N(K_i)) < |K_i|$$

Now let:

$$P := N(K_1) = A_1 \setminus y_1 \cup N(L_1) \quad Q := N(K_2) = A_1 \setminus y_2 \cup N(L_2)$$

where $L_1, L_2 \subseteq \{2, \dots, n\}$. Then we have:

$$P \cap Q = N(L_1 \cap L_2) \cup A_1 \setminus \{y_1, y_2\} \quad P \cup Q = N(L_1 \cup L_2) \cup A_1$$

We know:

$$r_{\mathcal{M}_Y}(P) = r_{\mathcal{M}_Y}(N(K_1)) < |K_1| = |L_1| + 1 \quad r_{\mathcal{M}_Y}(Q) = r_{\mathcal{M}_Y}(N(K_2)) < |K_2| = |L_2| + 1$$

and hence:

$$r_{\mathcal{M}_Y}(P) + r_{\mathcal{M}_Y}(Q) < |L_1| + |L_2| + 1$$

But we also know, by submodularity:

$$\begin{aligned} r_{\mathcal{M}_Y}(P) + r_{\mathcal{M}_Y}(Q) &\geq r_{\mathcal{M}_Y}(P \cap Q) + r_{\mathcal{M}_Y}(P \cup Q) \\ &= r_{\mathcal{M}_Y}(N(L_1 \cap L_2) \cup A_1 \setminus \{y_1, y_2\}) + r_{\mathcal{M}_Y}(N(L_1 \cup L_2) \cup A_1) \\ &\geq r_{\mathcal{M}_Y}(N(L_1 \cap L_2)) + r_{\mathcal{M}_Y}(N(L_1 \cup L_2) \cup A_1) \\ &\geq |L_1 \cap L_2| + |L_1 \cup L_2| + 1 \\ &= |L_1| + |L_2| + 1 \end{aligned}$$

Contradiction. ■

If we let $\mathcal{M}_Y$ be the free matroid on Y , then we get the Hall's marriage theorem:

Corollary 3. (Hall's Marriage Theorem, 1935) Given a bipartite graph $\Delta[X \cup Y]$, X can be perfectly matched to some subset of Y if and only if $\forall K \subseteq X$,

$$|N(K)| \geq |K|$$
■

Corollary 4. Given a bipartite graph $\Delta[X \cup Y]$, and a matroid structure $\mathcal{M}_Y$ on Y . Then $\exists$ a subset $X' \subseteq X$ of size $|X| - d$ that can be perfectly matched to some independent subset of Y if and only if $\forall T \subseteq X$,

$$r_{\mathcal{M}_Y}(N(T)) \geq |T| - d$$

Proof. We construct a new matroid $\mathcal{M}_Z$ where $Y \subseteq Z$, and a new bipartite graph $\Delta[X \cup Z]$ in this way: Let $Z = Y \cup \{e_1, \dots, e_d\}$, where each e_i is self-independent, i.e., a $U_{1,1}$, denoted as $\mathcal{U}$. Hence

$$\mathcal{M}_Z = \mathcal{M}_Y \oplus \mathcal{U}^{\oplus d}$$

Then, for $\Delta[X \cup \{e_1, \dots, e_d\}]$, make it a complete bipartite graph.

In this setting, by Theorem 2, we know that X can be perfectly matched to an independent

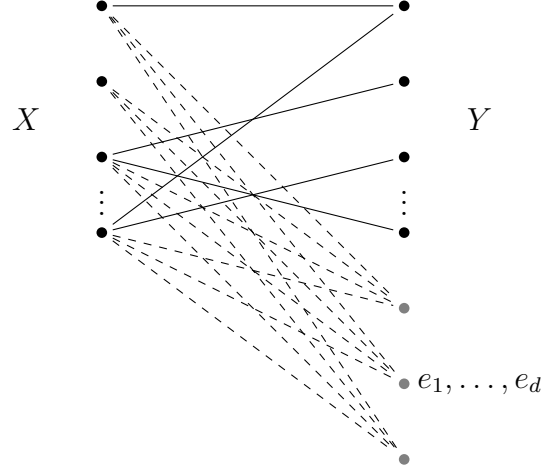


Figure 1: Bipartite graph $\Delta[X \cup Z]$ with $Z = Y \cup \{e_1, \dots, e_d\}$.

set in $\mathcal{M}_Z$ if and only if $\forall T \subseteq X$

$$r_{\mathcal{M}_Z}(N(T)) \geq |T|$$

Also, X can be matched to an independent set in $\mathcal{M}_Z$ if and only if $\exists X' \subseteq X$ of size $|X| - d$ that can be matched to an independent set in $\mathcal{M}_Y$. Because: $\Leftarrow$: Just let the remaining d elements in X match to $\{e_1, \dots, e_d\}$. $\Rightarrow$: Let the perfect matching be $\phi : X \rightarrow Z$. Then we delete $\phi^{-1}(\text{Im}\phi \cup \{e_1, \dots, e_d\})$ from X , the remaining still have $\geq |X| - d$ elements and can be matched to Y via ϕ . Therefore, it remains to show: for all $T \subseteq X$,

$$r_{\mathcal{M}_Y}(N(T)) \geq |T| - d \Leftrightarrow r_{\mathcal{M}_Z}(N(T)) \geq |T|$$

We know that since $\Delta[X \cup \{e_1, \dots, e_d\}]$ is complete, $\forall T \subseteq X$,

$$r_{\mathcal{M}_Z}(N(T)) = r_{\mathcal{M}_Y}(N(T)) + d$$

So we are done. ■

Corollary 4 gives us a way to get the rank function of the induced matroid $\Delta(\mathcal{M}_Y)$ on X :

Proposition 4. The rank function of $\Delta(\mathcal{M}_Y)$ is given by the following: For any $K \subseteq X$,

$$r_{\Delta(\mathcal{M}_Y)}(K) = \min\{r_{\mathcal{M}_Y}(N(I)) + |K - I| : I \subseteq K\}$$

Proof:

$$\begin{aligned}
r_{\Delta(\mathcal{M}_Y)}(K) &= \max_{J \subseteq K} \{|J| : J \text{ is independent in } \Delta(\mathcal{M}_Y)\} \\
&= \max_{J \subseteq K} \{|J| : J \text{ can be matched to some independent set of } \mathcal{M}_Y\} \\
&= \max_d \{|K| - d : r_{\mathcal{M}_Y}(N(I)) \geq |I| - d, \forall I \subseteq K\} \\
&= \max_d \{|K| - d : r_{\mathcal{M}_Y}(N(I)) + |K - I| \geq |K| - d, \forall I \subseteq K\} \\
&= \min\{r_{\mathcal{M}_Y}(N(I)) + |K - I| : I \subseteq K\}
\end{aligned}$$

Thus proves the statement in Theorem 1. ■

The above observations hint the definition of the matroid union:

Def 1. Suppose we have multiple matroids $\mathcal{M}_1 = (X, \mathcal{I}_1), \dots, \mathcal{M}_n = (X, \mathcal{I}_n)$ on the same ground set X . Then we can define a new matroid $\mathcal{M} = (X, \mathcal{I})$ where

$$\mathcal{I} = \left\{ \bigcup_{j=1}^n I_j \mid I_j \in \mathcal{I}_j, j \in [n] \right\}$$

Equivalently, $I \in \mathcal{I}$ if and only if I can be partitioned into disjoint subsets $J_1, \dots, J_n$ (possibly empty) s.t. $J_i \in \mathcal{I}_i, i \in [n]$. We denote this matroid as $\mathcal{M} := \bigvee_{i=1}^n \mathcal{M}_i$.

Why is $\mathcal{M}$ a well-defined matroid? One can check the satisfaction of the axioms. But here we use the Rado matroid to characterize it: Consider the bipartite graph $\Delta[X \cup Y]$ where Y is just a n copies of X , say $Y = \sqcup_{i=1}^n X_i$ where $X_i = X$. And for each $x \in X$ and every corresponding $x_i \in X_i, i \in [n]$, we add an edge (x, x_i) . Here is an example for $n = 2$:

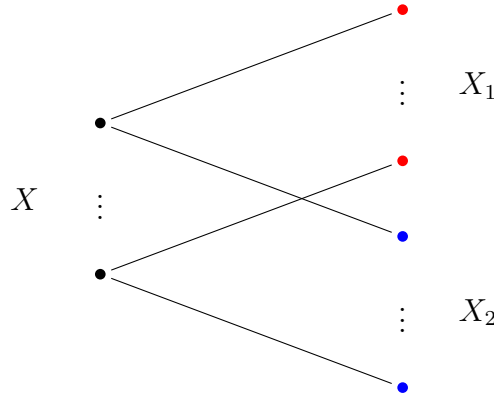


Figure 2: Bipartite graph $\Delta[X \cup Y]$ with $Y = X_1 \cup X_2$, where each X_i is a copy of X

Now we give a matroid structure on Y , which is the direct sum of the X_i s, i.e., let $\mathcal{M}_Y = \mathcal{M}_1 \oplus \dots \oplus \mathcal{M}_n$. Then we claim that the induced matroid $\Delta(\mathcal{M}_Y)$ on X is the matroid $\mathcal{M}$ we want. In other words:

$$\Delta\left(\bigoplus_{i=1}^n \mathcal{M}_i\right) = \bigvee_{i=1}^n \mathcal{M}_i$$

Proof: I is independent in $\Delta(\mathcal{M}_Y) \Leftrightarrow I$ can be perfectly matched to an independent set, say $J = \sqcup_{i=1}^n J_i$ in $\mathcal{M}_Y$, where each $J_i \in \mathcal{I}(\mathcal{M}_i)$. By our construction of Δ , this is equivalent of saying: I can be partitioned into $\sqcup_{i=1}^n I_i$, where each I_i is exactly J_i . ■

One immediate benefit of this characterization is: we can write down the rank function naturally. For any $S \subseteq X$, by our main theorem:

$$\begin{aligned} r_{\mathcal{M}}(S) &= \min_{T \subseteq S} \{r_{\mathcal{M}_Y}(N(T)) + |S - T|\} \\ &= \min_{T \subseteq S} \left\{ \sum_{i=1}^n r_{\mathcal{M}_i}(T) + |S - T| \right\} \end{aligned}$$

■

References

[Oxl06] James G Oxley. *Matroid theory*, volume 3. Oxford University Press, USA, 2006.